ECON 3200 Discussion Note - Sep 2, 2026
Review: Returns to Scale

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1 What are returns to scale?

Suppose output is produced according to

$$Q = F(L, K),$$

where L is labor and K is capital.

Returns to scale asks:

> What happens to output if we increase **all** inputs by the same proportion?

For example, suppose we double both labor and capital:

$$(L, K) \to (2L, 2K).$$

We compare

$$F(2L, 2K)$$

with

$$2F(L, K).$$

More generally, if all inputs are multiplied by some factor $\lambda > 1$, compare

$$F(\lambda L, \lambda K)$$

with

$$\lambda F(L, K).$$

2 Constant, increasing, and decreasing returns to scale

Constant returns to scale

A production function has **constant returns to scale (CRS)** if

$$\boxed{F(\lambda L, \lambda K) = \lambda F(L, K)}$$

for any $\lambda > 0$.

If all inputs double, output doubles. If all inputs triple, output triples.

For example, if

$$F(L, K) = 100,$$

and

$$F(2L, 2K) = 200,$$

the production technology has constant returns to scale.

Increasing returns to scale

A production function has **increasing returns to scale (IRS)** if

$$\boxed{F(\lambda L, \lambda K) > \lambda F(L, K)}$$

for $\lambda > 1$.

If all inputs double, output more than doubles.

For example,

$$F(L, K) = 100,$$

but

$$F(2L, 2K) = 250.$$

Possible reasons of increasing returns include specialization, fixed costs, or technologies that become more productive at larger scale.

Decreasing returns to scale

A production function has **decreasing returns to scale (DRS)** if

$$\boxed{F(\lambda L, \lambda K) < \lambda F(L, K)}$$

for $\lambda > 1$.

If all inputs double, output less than doubles.

For example,

$$F(L, K) = 100,$$

but

$$F(2L, 2K) = 180.$$

Possible reasons include coordination or management difficulties as the scale of production becomes larger.

3 One-input production

Suppose labor is the only input:

$$Q = F(L).$$

Returns to scale now asks what happens to output when labor is scaled up.

Constant returns to scale

$$F(\lambda L) = \lambda F(L).$$

A simple example is

$$Q = 4L.$$

If $L = 10$,

$$Q = 40.$$

If labor doubles to $L = 20$,

$$Q = 80.$$

Therefore,

$$F(2L) = 2F(L),$$

so the technology has constant returns to scale.

Increasing returns to scale

Consider

$$Q = L^2.$$

If labor doubles,

$$Q' = (2L)^2 = 4L^2 = 4Q.$$

Output more than doubles, so the technology has increasing returns to scale.

Decreasing returns to scale

Consider

$$Q = L^{1/2}.$$

If labor doubles,

$$Q' = (2L)^{1/2} = \sqrt{2}\,Q.$$

Since

$$\sqrt{2} < 2,$$

output less than doubles. The technology has decreasing returns to scale.

4 One input: returns to scale and marginal product

Suppose

$$F(\lambda L) = \lambda F(L).$$

Set the reference level of labor equal to one and choose $\lambda = L$. Then

$$F(L) = LF(1).$$

Let

$$A = F(1).$$

Then

$$\boxed{Q = AL.}$$

This is a linear production function.

The marginal product of labor is

$$MPL = \frac{dQ}{dL} = A.$$

Therefore,

$$\boxed{\text{With one input, constant returns to scale implies a constant marginal product.}}$$

In this one-input case, the following statements are equivalent:

$$\boxed{\text{CRS} \iff \text{linear production function} \iff \text{constant MPL.}}$$

5 Two-input production

The connection between CRS and constant marginal product is special to the one-input case.

Suppose output depends on labor and capital:

$$Q = F(L, K).$$

Constant returns to scale requires

$$F(\lambda L, \lambda K) = \lambda F(L, K).$$

The key point is that **all inputs must be scaled together**.

Example: Cobb–Douglas production

Consider

$$Q = L^{1/2}K^{1/2}.$$

If both inputs double,

$$Q' = (2L)^{1/2}(2K)^{1/2}.$$

Then

$$Q' = 2L^{1/2}K^{1/2} = 2Q.$$

Therefore, this production function has constant returns to scale.

But the marginal product of labor is not constant

The marginal product of labor is

$$MPL = \frac{\partial Q}{\partial L} = \frac{1}{2}L^{-1/2}K^{1/2}.$$

Holding K fixed, MPL falls as L increases.

Therefore,

$$\boxed{\text{CRS does not generally imply constant marginal products.}}$$

The reason is that returns to scale and marginal product ask different questions.

Concept	**What changes?**	**Question**
Marginal product of labor	Labor changes; other inputs are fixed	How much does output change when labor increases?
Returns to scale	All inputs change by the same proportion	What happens to output when the entire scale of production increases?

6 Connection to the Ricardian model

In the Ricardian model used in lecture, labor is the only factor of production in each sector.

For cloth,

$$Q_C = MPL_C L_C,$$

and for wheat,

$$Q_W = MPL_W L_W.$$

The marginal products MPL_C and MPL_W are assumed to be constant.

For example, if

$$MPL_C = 2, \qquad MPL_W = 4,$$

then

$$Q_C = 2L_C, \qquad Q_W = 4L_W.$$

Why is this constant returns to scale?

For cloth,

$$Q_C = MPL_C L_C.$$

If labor in the cloth sector is multiplied by λ,

$$Q'_C = MPL_C(\lambda L_C).$$

Therefore,

$$Q'_C = \lambda MPL_C L_C = \lambda Q_C.$$

So

$$\boxed{Q_C = MPL_C L_C \text{ exhibits constant returns to scale.}}$$

The same argument applies to wheat.

Constant MPL and CRS in the Ricardian model

Because each sector has only one input, the Ricardian model gives us the special one-input equivalence:

$$\boxed{\text{constant MPL} \iff \text{linear production function} \iff \text{constant returns to scale.}}$$

This is why output is proportional to the number of workers employed in each sector.

7 Why does constant MPL matter for the PPF?

Because MPL_C and MPL_W are constant, moving one worker from cloth to wheat always has the same effect:

- wheat output rises by MPL_W;
- cloth output falls by MPL_C.

Therefore, the opportunity cost of wheat in terms of cloth is constant:

$$\boxed{\frac{MPL_C}{MPL_W}.}$$

Starting from the labor constraint,

$$L = L_C + L_W,$$

and using

$$L_C = \frac{Q_C}{MPL_C}, \qquad L_W = \frac{Q_W}{MPL_W},$$

we obtain

$$L = \frac{Q_C}{MPL_C} + \frac{Q_W}{MPL_W}.$$

Rearranging,

$$\boxed{Q_C = MPL_C L - \frac{MPL_C}{MPL_W} Q_W.}$$

The PPF is therefore a straight line with slope

$$-\frac{MPL_C}{MPL_W}.$$

The logic is:

$$\boxed{\text{constant MPL} \Rightarrow \text{linear production technologies} \Rightarrow \text{constant opportunity cost} \Rightarrow \text{linear PPF.}}$$

Summary

Returns to scale asks what happens when **all inputs are scaled together**:

$$F(\lambda L, \lambda K) \begin{cases} = \lambda F(L,K), & \text{CRS,} \\ > \lambda F(L,K), & \text{IRS,} \\ < \lambda F(L,K), & \text{DRS.} \end{cases}$$

Do not confuse returns to scale with marginal product:

$$\boxed{\text{Returns to scale: change all inputs proportionally.}}$$

$$\boxed{\text{Marginal product: change one input while holding other inputs fixed.}}$$

In the Ricardian model, labor is the only input and

$$Q = MPL \cdot L, \qquad MPL = \text{constant.}$$

Therefore,

$$\boxed{\text{constant MPL} \Longleftrightarrow \text{constant returns to scale}}$$

in this one-input setting.

Constant marginal products also imply a constant opportunity cost and therefore a linear production possibilities frontier.